JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU& RTMNU

Education to Eternity

॥ज्ञानम् सर्वार्थ साधनम्॥

B.Tech / M.Tech / MBA / MCA Summer / Winter - 20
(ESE / Make-up / Supplementary) Examination

Branch: CS/AI/DS

Subject:Linear Algebra & Complex Analysis **Sem:-I**

Subject Code: (CS/AI/DS2T001) **Marks: 100**

Date of Exam: / /20 **Time: 3Hrs.**

Instructions:

1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**
4. **Use of Programmable Calculator is prohibited.**
5. **Read the question paper carefully (Branch,Semester,Scheme)before attempting the questions.**

COs: Students should be able to:

1. CO1: Describe concept of Advance linear Algebra, ODE, Statistics, Complex numbers, Function of Complex variables.

CO2: Illustrate the concept of Advance linear Algebra, ODE, Statistics, Complex numbers, Function of Complex variables by using examples.

CO3: Apply the knowledge of Advance linear Algebra, ODE, Statistics, Complex numbers, Function of Complex variables to solve the engineering problems.

CO4: Analyze the problems and results of Advance linear Algebra, ODE, Statistics, Complex numbers, Function of Complex variables to solve the real world problems.

CO5: Evaluate the problems by using Advance linear Algebra, ODE, Statistics, Complex numbers, Function of Complex variables.

CO6: Create the methods or model by using Advance linear Algebra, ODE, Statistics, Complex numbers, Function of Complex variables

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Marks	Bloom's Level and CO
Q1	Recall statement of Cayley-Hamilton theorem.	2	L1/CO1
Q2	Find determinant of matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$	2	L1/CO1
Q3	Classify the Cauchy-Riemann necessary condition.	2	L2/CO2
Q.4	Write standard Linear differential equation and its solution.	2	L1/CO1
Q.5	Find the statement and formula of coefficient of correlation 'r'.	2	L1/CO1
Q6	Find the equation of the regression line of y on x.	2	L1/CO1
Q7	DefineModulus &Amplitude of complex number.	2	L1/CO1
Q8	Define analytic function.	2	L1/CO1
Q9	Write down the statement of Cauchy's theorem.	2	L1/CO1
Q.10	If $u = y^3 - 3x^2y$ find u is harmonic or not?	2	L2/CO2
Q.11	Define Eigen vector.	2	L1/CO1
Q.12	Tell the formula for the rank correlation coefficient for non-repeated ranks.	2	L2/CO2

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Subject:Linear Algebra & Complex Analysis **Marks: 100**

Subject Code: (CS/AI/DS2T001) **Time: 3Hrs.**

Date of Exam: 01/06/2024

SECTION B

1. **All Questions are compulsory.**

Q.13 Analyze the Eigen values of the matrices $A = \begin{bmatrix} 5 & 7 & -5 \\ 0 & 4 & -1 \\ 2 & 8 & -3 \end{bmatrix}$ 4 L4/CO4

Q.14 Solve $(x+1)\frac{dy}{dx} - 2y = (x+1)^4$ 4 L3/CO3

Q.15 Using the following data to fit a curve $y = ax + bx^2$ 4 L 4/CO4

x	-2	-1	0	1	2
y	-72	-46	-12	35	93

Q.16 Show that$(a+ib)^{m/n} + (a-ib)^{m/n} = 2(a^2+b^2)^{m/2n} cos\frac{m}{n} tan^{-1}\frac{b}{a}$ 4 L3/CO3

Q.17 Show that$u = y^3 - 3x^2y$ is Harmonic function. 4 L3/CO3

Q.18 Analyze the rank of the following matrices $A = \begin{bmatrix} 1 & 2 & -4 & 5 \\ 2 & -1 & 3 & 6 \\ 8 & 1 & 9 & 7 \end{bmatrix}$ 4 L4/CO4

$R_2 = R_2 - 2R_1$ $R_3 = R_3 - 8R_1$

SECTION C

1. **Solve Any FOUR Questions.**

Q.19 Verify Cayley-Hamilton theorem for the matrixhence find A^{-1}

$A = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}$ 7 L4/CO4

Q.20 Solve $(1+x^2)\frac{dy}{dx} + y = e^{tan^{-1}x}$ 7 L6/CO6

Q.21 Analysethe rank correlation coefficient for the following data. 7 L4/CO4

x	2	4	5	6	8	11
y	18	12	10	8	7	5

Q.22 If $\tan(A+iB) = x + iy$ then prove that 7 L5/CO5

1. $tan2A = \frac{2x}{1-x^2-y^2}$, 2. $tanh2B = \frac{2y}{1+x^2+y^2}$

Q.23 Analyse$\oint_c \frac{3z^2+z}{z^2-1} dz$ where 'c'is circle $|z-1| = 1$ 7 L4/CO4

Q.24 Solve $(x^2y^2+2)y\,dx + (2-2x^2y^2)x\,dy = 0$ 7 L6/CO6

Q24- $(x^2y^2+2)y\,dx + (2-2x^2y^2)x\,dy = 0$

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SECTION D

1. **Solve any TWO Questions**

Q.25 For matrix A= $\begin{bmatrix} 9 & -1 & 9 \\ 3 & -1 & 3 \\ -7 & 1 & -7 \end{bmatrix}$ construct the model matrix B.Prove that $B^{-1}AB$ is a diagonal matrix. 12 L6/CO6

Q.26 Evaluate the rank of correlation coefficient for the following data. 12 L5/CO5

x	68	64	75	50	64	80	75	40	55	64
y	62	58	68	45	81	60	68	48	50	70

Σ 635
Σ 655

Q.27 Evaluate all the values of $x^{12} - 1 = 0$ 12 L5/CO5

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